Day 1: Smart Parking Management System

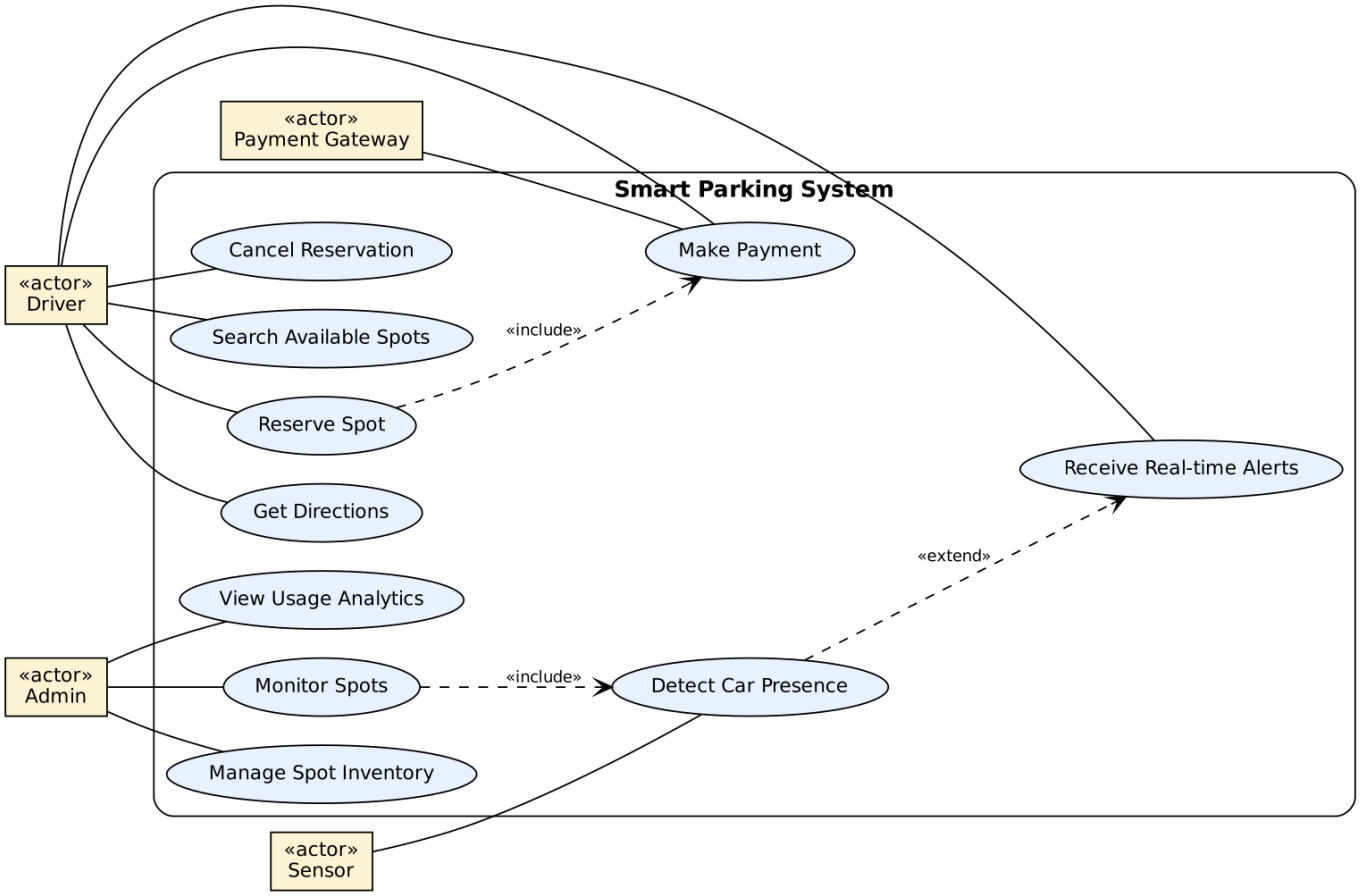
Course: Software Engineering and System Design (24CS301PC311) | Assignment 2: UML Modeling

Case explanation

A Smart Parking System lets drivers use a mobile app to find and reserve parking spots. Drivers check real-time availability, book a spot, pay online and receive directions to the lot. Sensors installed at each spot detect whether a car is present and push live updates, so the app and the drivers' alerts always reflect the true state. Admins manage the spot inventory and monitor usage through analytics. The five diagrams below model this system from different views: its functions (use case), structure (class), interactions (sequence), workflow (activity) and spot life-cycle (state machine).

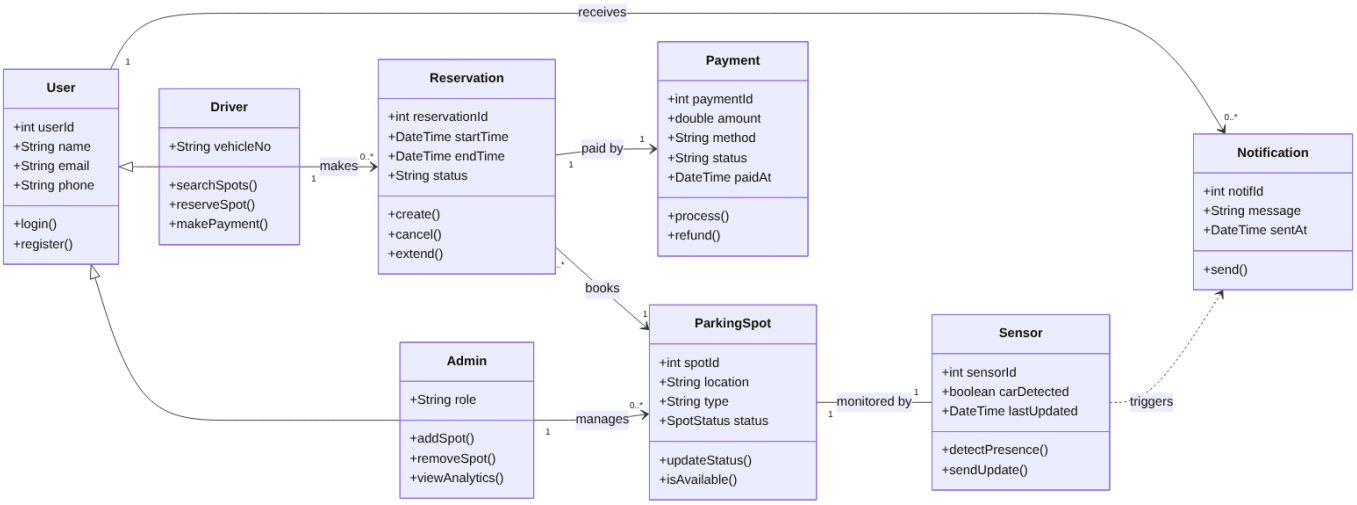
1. Use Case Diagram

Actors: Driver, Admin, Sensor and the external Payment Gateway. Reserving a spot includes making payment; monitoring includes detecting presence, which can trigger an alert.



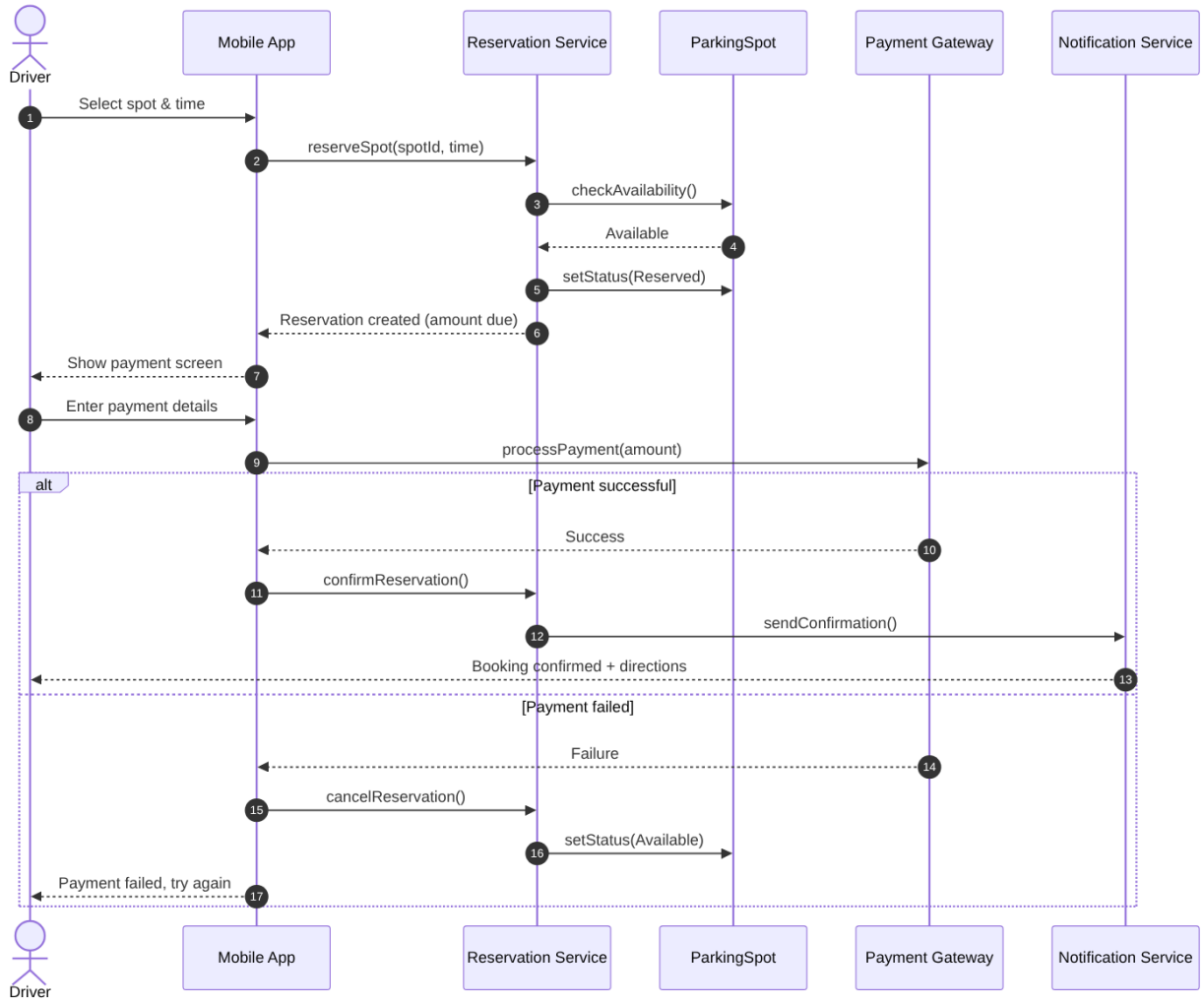
2. Class Diagram

Driver and Admin specialise User. A Driver makes Reservations for ParkingSpots; each Reservation has one Payment; each ParkingSpot is monitored by one Sensor, which triggers Notifications.



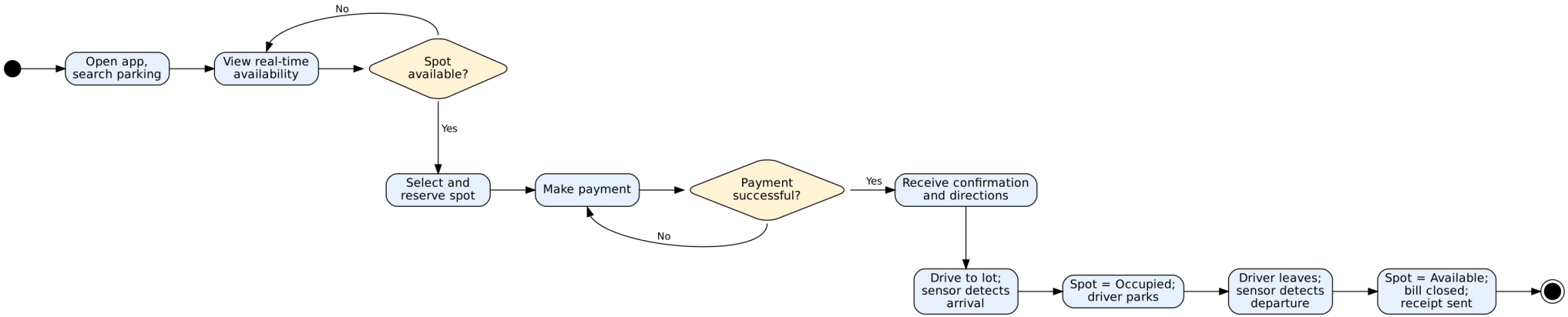
3. Sequence Diagram: Reserve a spot and pay

The driver selects a spot, the system checks availability and marks it Reserved, then payment is processed. On success the booking is confirmed; on failure the reservation is cancelled and the spot is released.



4. Activity Diagram: Search to exit

Workflow from searching and reserving, through payment and arrival, to parking, leaving and receipt generation.



5. State Machine Diagram: ParkingSpot

Available → Reserved → Occupied → Available, plus cancellation/timeout back to Available and an Out of Service state for faults.

